

Econ 30065: Microeconomics

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Lecture 15: Competitive Market Equilibrium

Competitive markets

- In a perfectly competitive market, buyers and sellers are **price takers**
- Characteristics of perfectly competitive markets:
 - Buyers and sellers face no **transaction costs**
 - Buyers can easily purchase from the seller who offers a product at the best terms
 - Products are **homogeneous**
 - Price from different sellers must be the same if they both want to make sales
 - There are **many buyers and sellers**, each accounting for a small fraction of the overall demand or supply of the good
 - A single producer can increase or reduce its supply, without affecting the market price
 - A single consumer can increase or reduce its demand, without affecting the market price

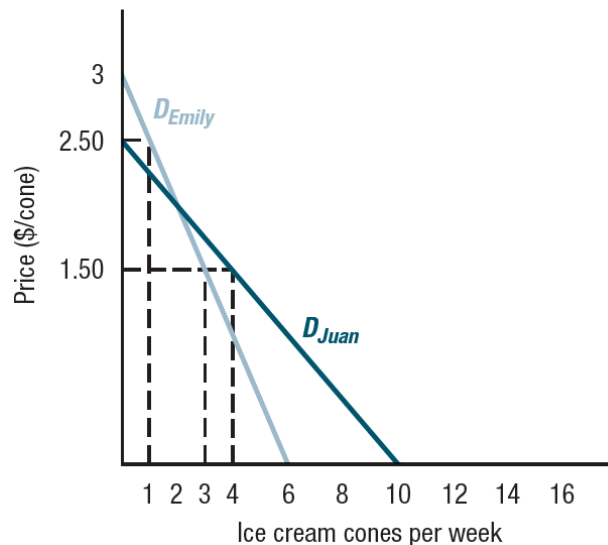
Markets are really perfectly competitive?

- Few markets meet those conditions:
 - transactions costs: transport costs (travel or delivery), travelling to another town etc, searching for the best price
 - Products differ, at least a little bit
 - Many market don't have lots of firms
- Nevertheless, useful to study perfectly competitive markets for 2 reasons:
 - Many markets are highly competitive, behave similarly to perfectly competitive markets
 - Useful benchmark for market efficiency

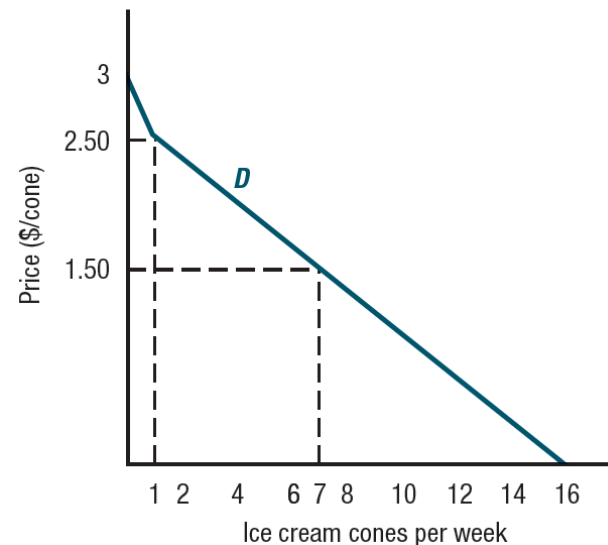
Market Demand

- **Market demand (at a given price):** sum of the demands of all the individual consumers
- **Market demand curve:** horizontal sum of the individual demand curves

(a) Individual demand



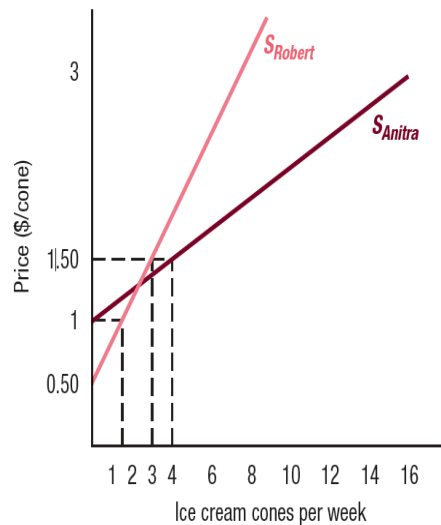
(b) Market demand



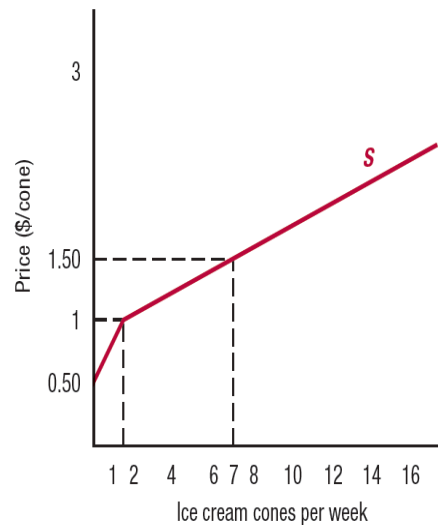
Market Supply

- **Market supply (at a given price):** sum of the supply of all the individual sellers
- **Market supply curve:** horizontal sum of the individual supply curves

(a) Individual supply

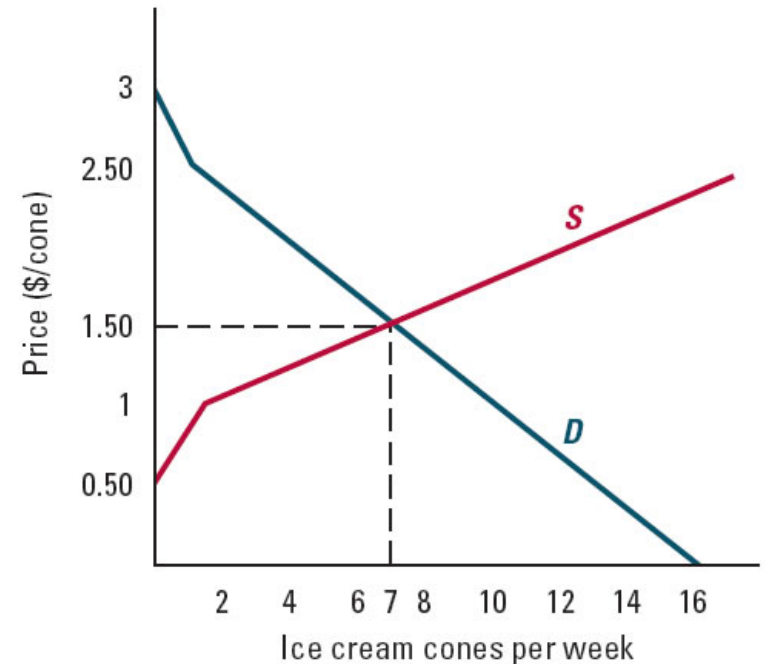


(b) Market supply



Competitive market equilibrium

- At equilibrium price, $Q_s = Q_d$
=> Market clears
- Given demand and supply functions, use algebra to find the equilibrium



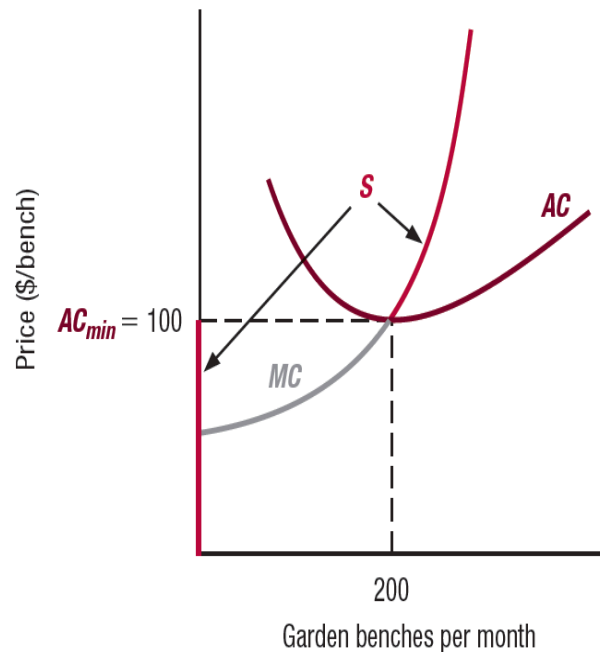
Short-Run and Long-Run Supply Curves

- Short-Run and Long-Run supply curves may be different for 2 reasons:
 - Fixed vs. variable inputs
 - Different set of firms
- Market supply curves:
 - **Short-run supply:** Add up the short-run supply curves of all *currently active* firms (they start at AVC_{min})
 - **Long-run supply:** Add up the long-run supply curves of all *potential* suppliers (they start at AC_{min})
 - **Long-run supply with *free entry*:** anyone who wishes to start a firm has free access to the same technology and entry is unrestricted (unlimited number of potential firms)
 - Long-run market supply curve is a horizontal line at AC_{min}

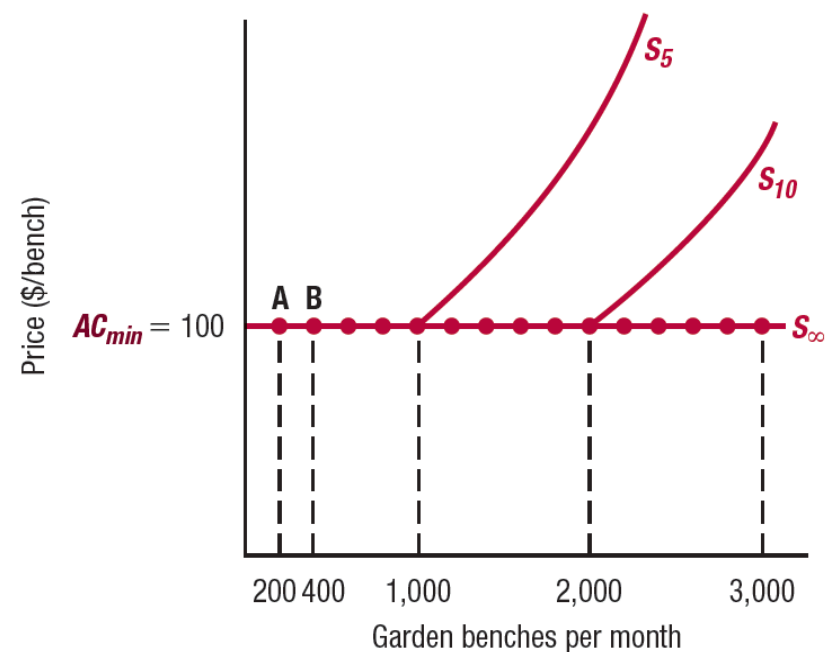
The Long -Run Market Supply Curve with Free Entry

In the LR, firms in a perfectly competitive market MUST operate at their efficient scale of production (where $AC=MC$)

(a) Individual supply



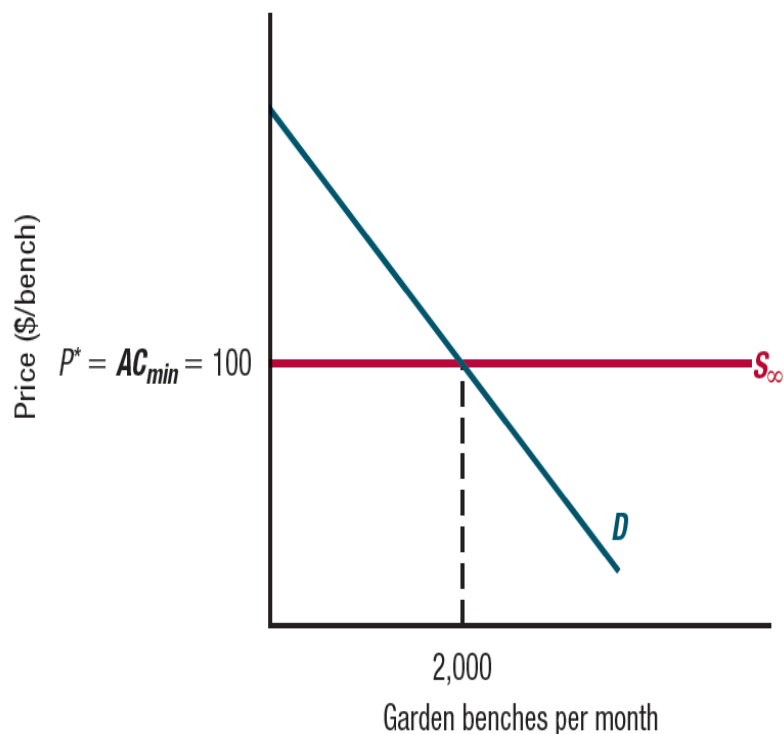
(b) Market supply



Long-Run Competitive Equilibrium with Free Entry

➤ Properties:

- Firms earn zero profit
 - Otherwise if positive profits new firms would enter
 - If negative profits some firms would exit
 - Until $P = AC_{min}$
- Equilibrium price equals AC_{min}
- Each active firm produces at its efficient scale of production



Social welfare

- Beyond determining equilibrium price and quantity, it is important to understand and measure changes in social **welfare** when prices and quantities consumed change:

What are the gains and losses in monetary terms?

- To compute them:
 - Consumer Surplus
 - Producer surplus
 - Aggregate surplus

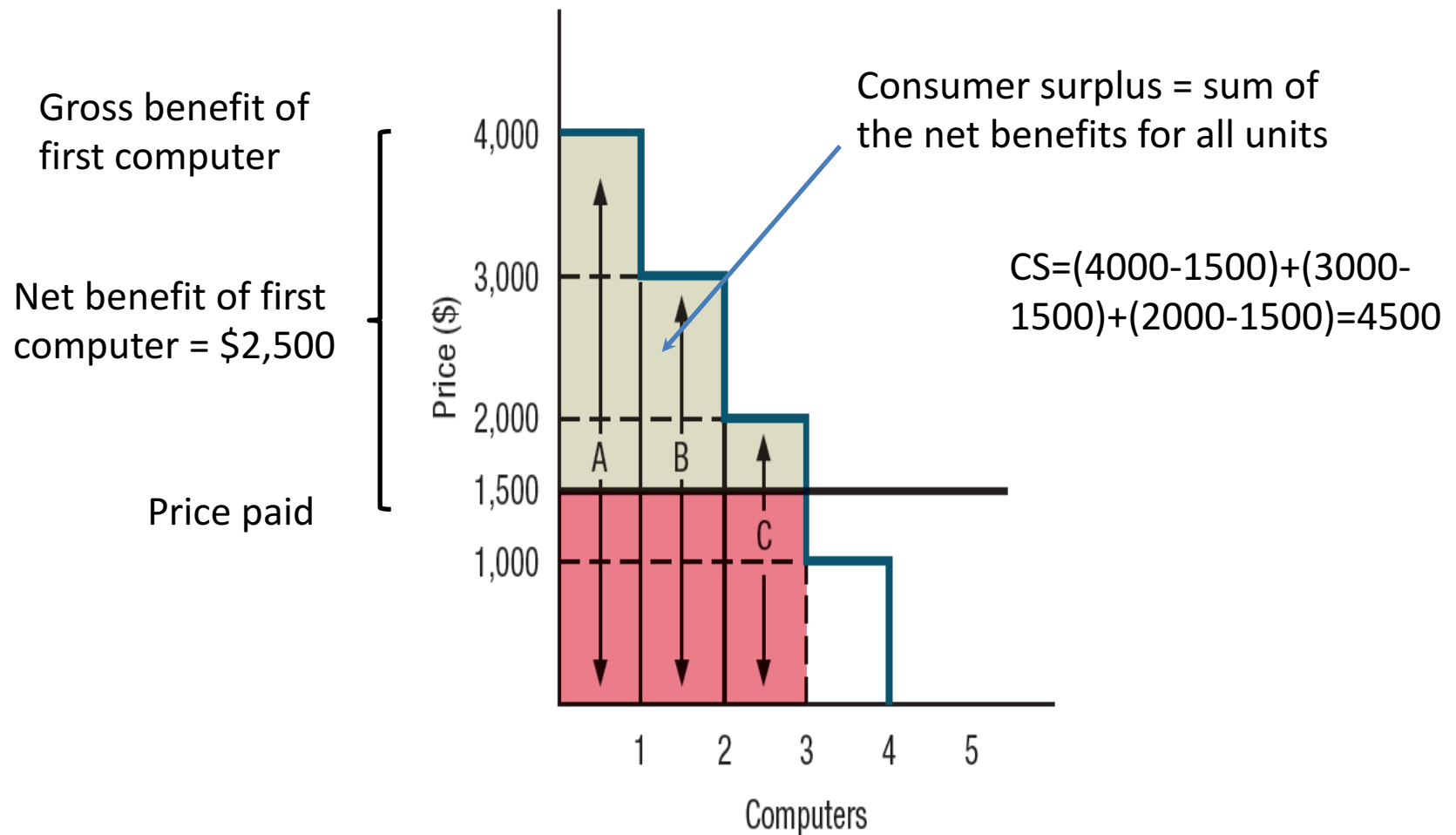
Consumer Surplus

- **Consumer surplus:** net benefit that a consumer receives from participating in the market for some good:

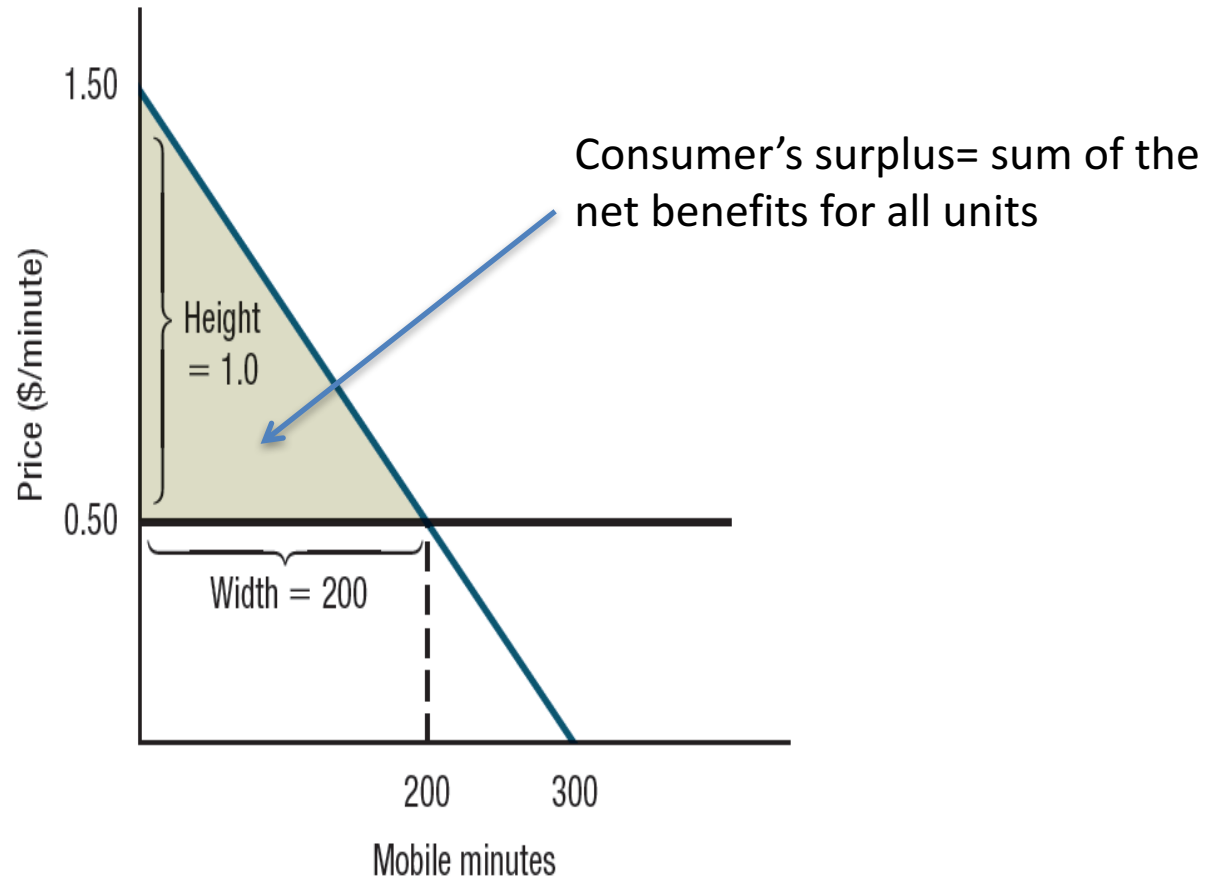
Sum of (how much he is willing to pay each unit of a good – price he actually pays for the good)

- Demand tells us how much he is willing to pay each unit of a good
- Market price tells us how much he actually pays for the good

Measuring Consumer Surplus (discrete units)



Measuring Consumer Surplus



Consumer's Surplus formula

Area of the triangle below demand and above market price:

$$CS = \frac{Q(Y \text{ intercept} - P)}{2}$$

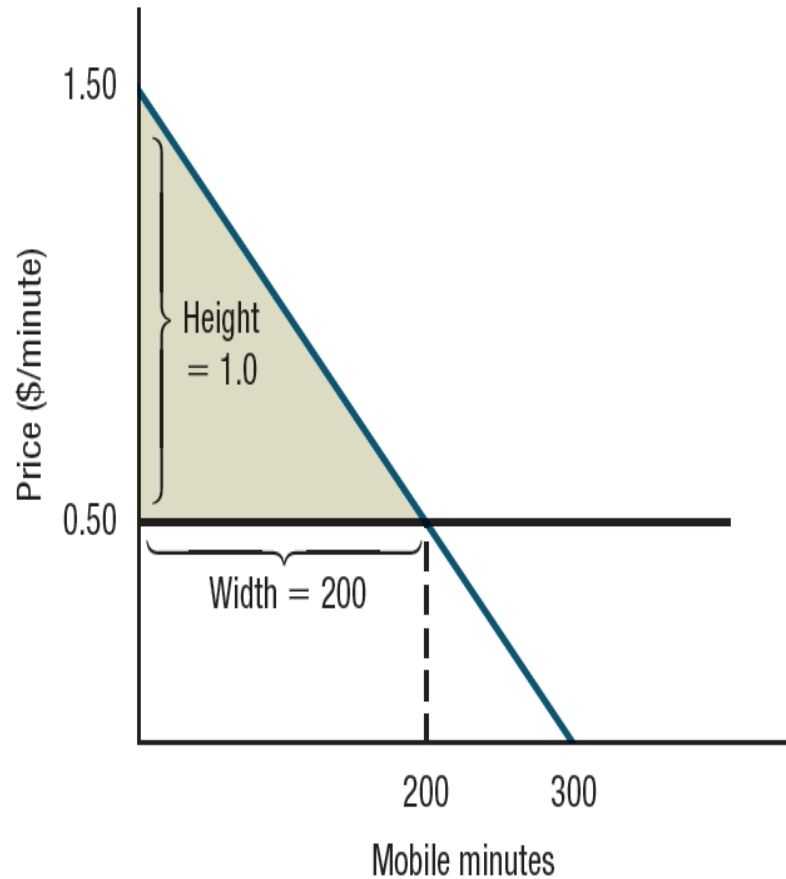
Example: Measuring Consumer Surplus

Abigail's monthly demand for minutes of mobile phone service is:

➤ $W = 300 - 200P_w$

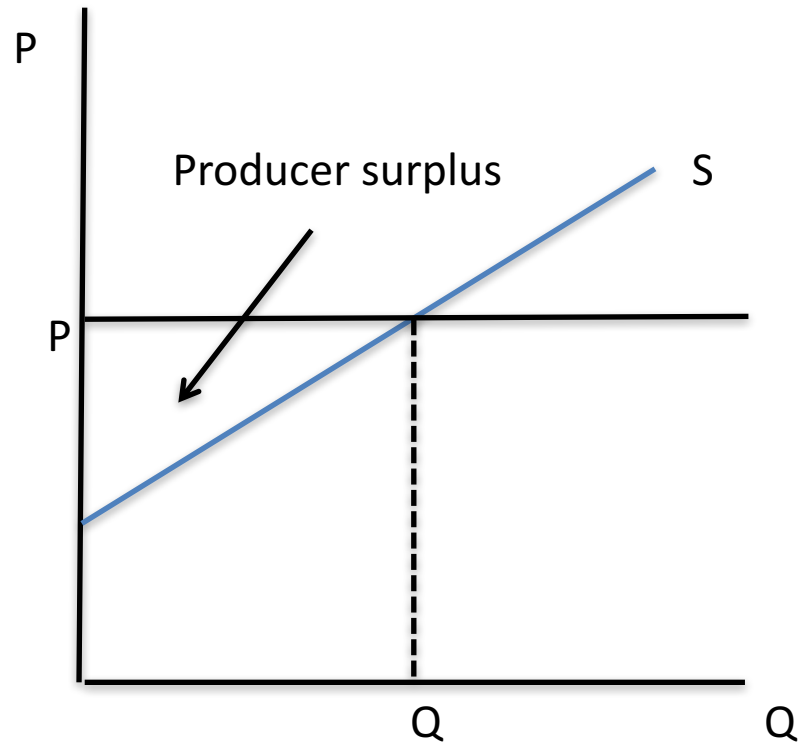
➤ $P_w = 0.5$

$$CS = \frac{200(1.5 - 0.5)}{2} = 100$$



Producer Surplus

- **Producer Surplus** =
revenue – variable costs
 - Area above the supply
and below the price, up
to Q: $0.5(P - S(Q=0))Q$
- **Profit** = producer surplus
– fixed costs
- Fixed costs (sunk costs)
ignored in taking
economic decisions



Aggregate Surplus

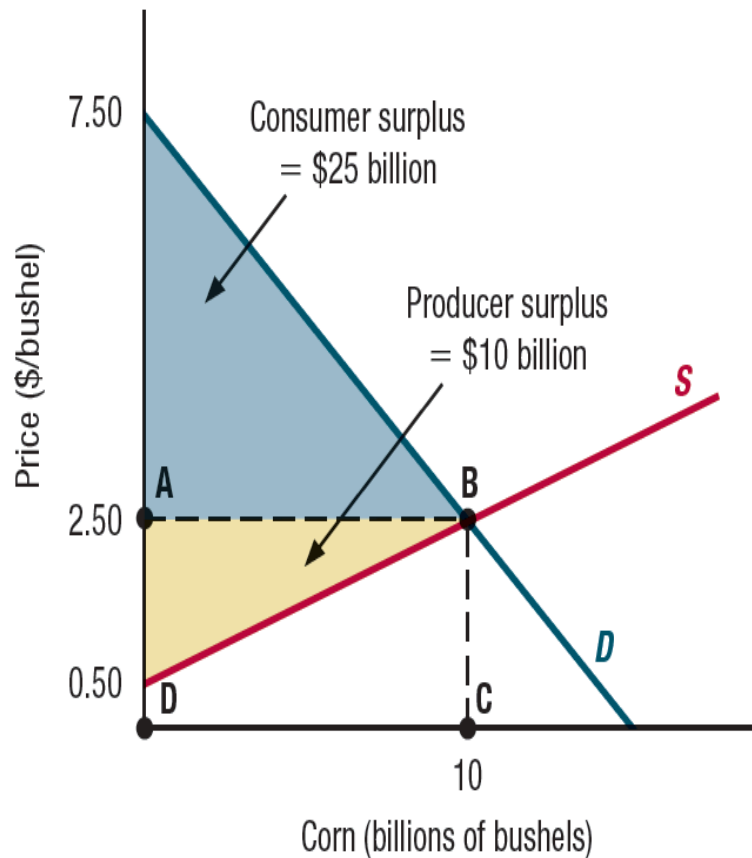
- **Aggregate surplus:** captures the net benefit created by the production and consumption of a good:
- **Aggregate surplus = aggregate CS + aggregate PS**

Consumers and Producers Surplus

- ***(Aggregate) Consumer surplus*** is the sum of consumers' total willingness to pay less their total expenditure ($P \times Q$)
 - Sum of individual consumers' surpluses
 - Also called simply consumer surplus
- ***(Aggregate) Producer surplus*** is the sum of firms' revenues ($P \times Q$) less avoidable costs
 - Sum of individual firms' producer surpluses
 - Also called simply producer surplus
- **Aggregate surplus = Consumer surplus + Producer Surplus**

Consumers and Producers Surplus

(b) Consumer and producer surplus

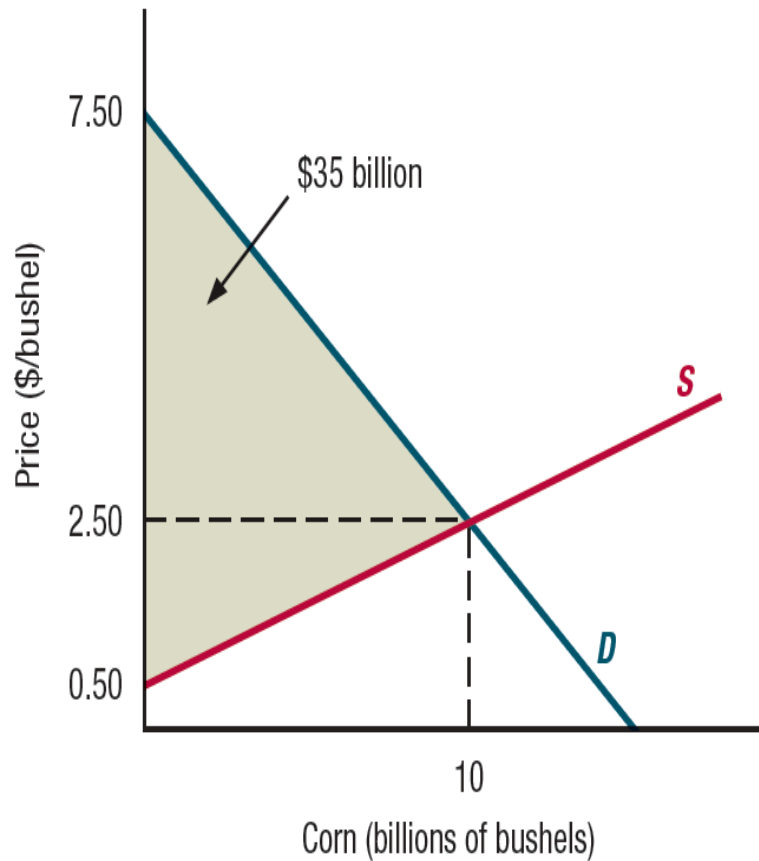


$$CS = 0.5(\text{Demand } Y_{\text{intercept}} - \text{price})Q$$

$$PS = 0.5(\text{Price} - \text{Supply } Y_{\text{intercept}})Q$$

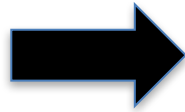
Aggregate Surplus

(a) Aggregate surplus



Economic Efficiency

- **Pareto efficient (or Pareto optimal):** an economic outcome at which it is impossible to make anyone better off without making someone else worse off



At the efficient outcome the aggregate surplus is maximized

Perfectly competitive markets and efficiency

- Perfectly competitive market produces an outcome that is economically efficient

Deadweight Loss

- **Deadweight Loss:** a reduction in aggregate surplus below its maximum possible value